

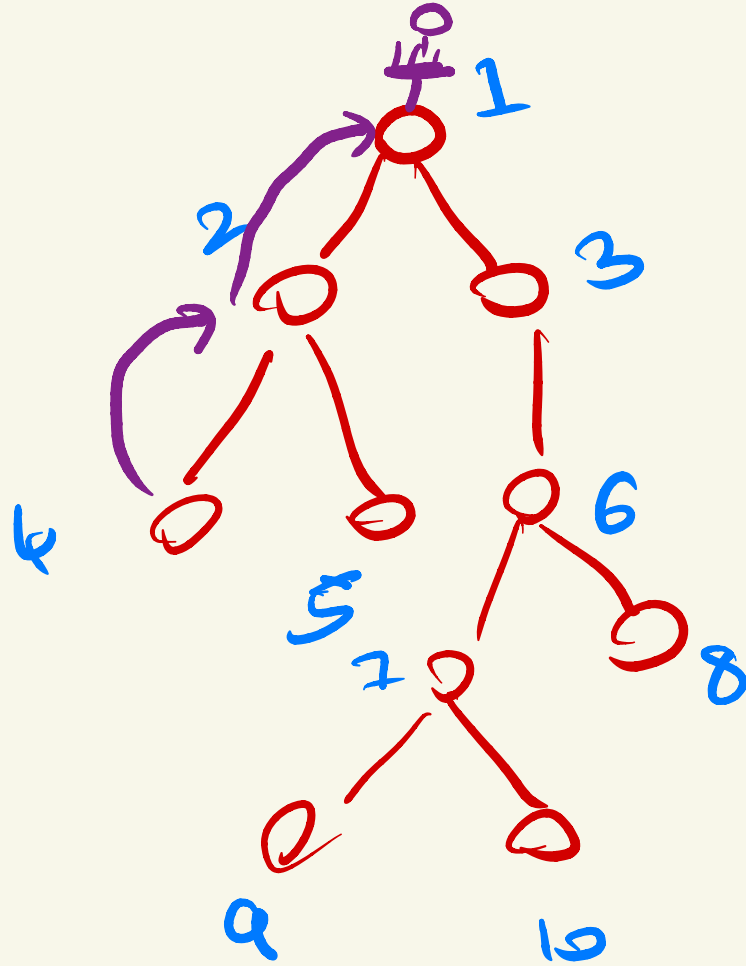
Trees 3

Binary Lifting

LCA

- Find kth Parent of any node in a Tree [Problem Link](#)
- Find LCA of 2 nodes [Problem Link](#)
 - $O(\log n * \log n)$
 - Precomputation of logs $\rightarrow O(\log n)$

Distance Queries



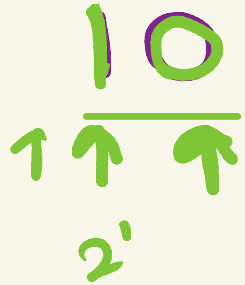
2
 2 k
 4 2 → 2
 4 3 → -1


$$\text{par}[2] = 1$$

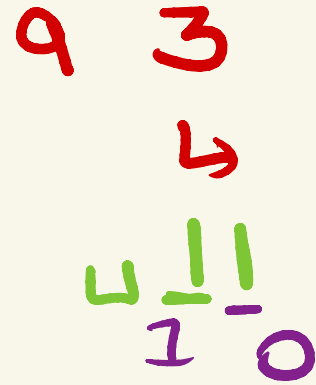
4 2

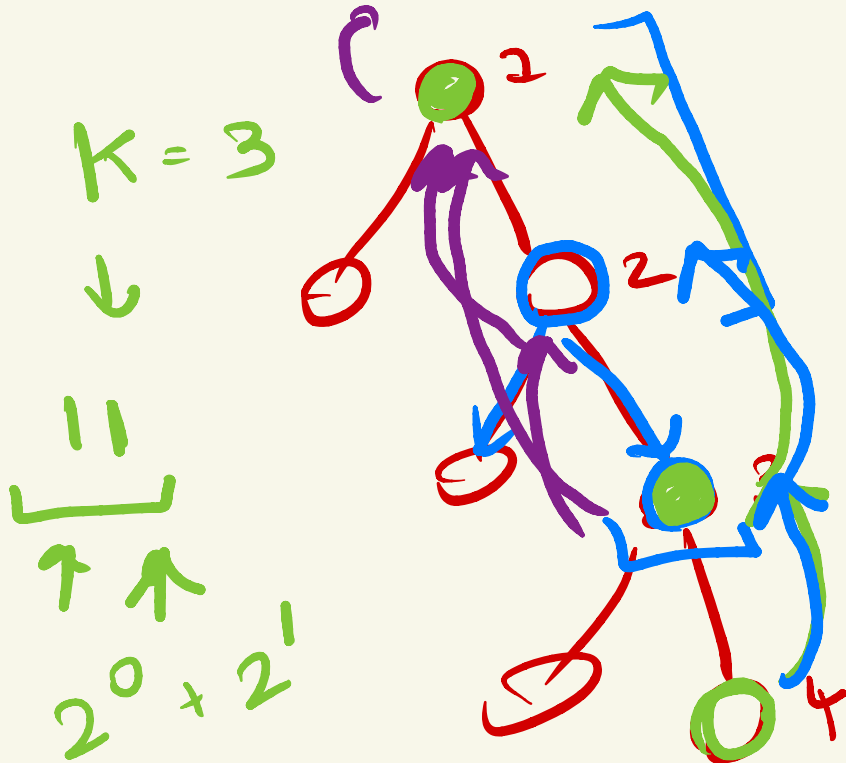
$$x = \text{par}[x]$$

Made with Goodnotes

$$\log n$$


2nd 4th 8th ...





1	0	0	0
2	1	0	0 0
3	2	1	0 -
	<u>0</u>	1	2

$dfs(node, par)$

1st parent of 1st por
of 3

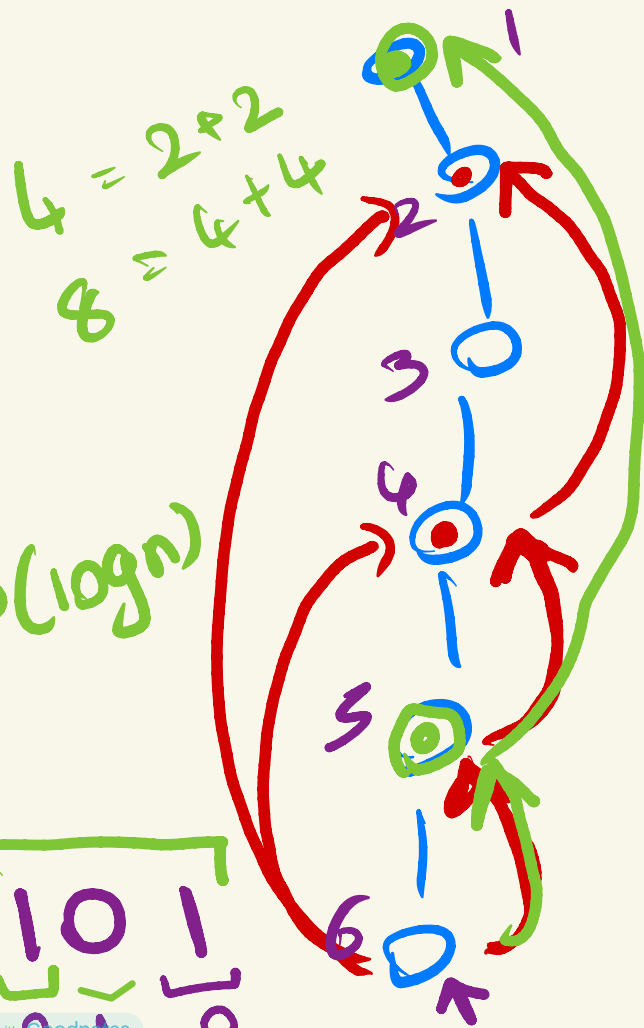
DFS

$\log_2 n$ parents

$$\text{par}[\text{node}][\underline{i}] = O((n+q)\log n)$$

$$\text{par}[\text{par}[\text{node}][\underline{i-1}]][\underline{i-1}]$$

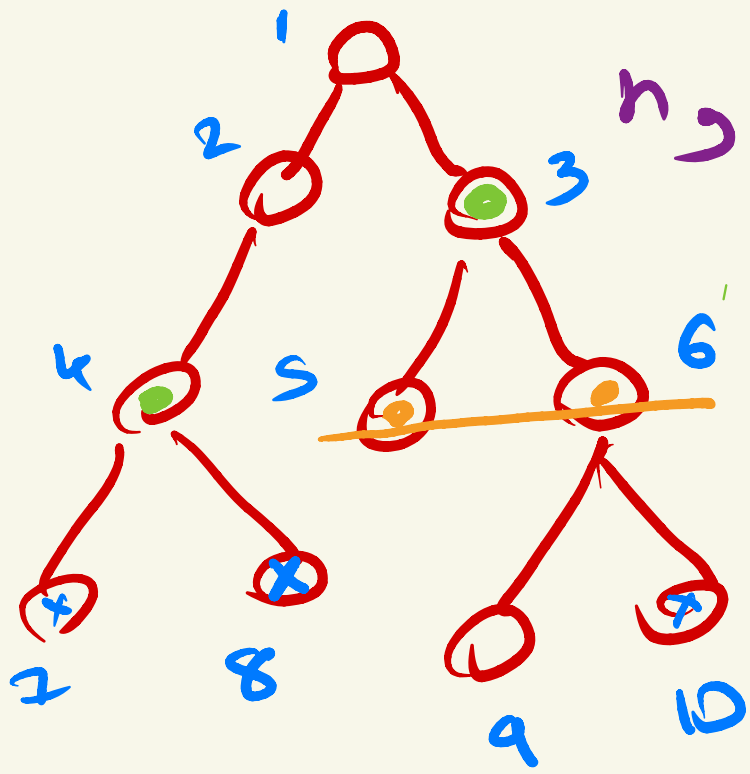
$$O(n \log n + q \log n)$$



$$\text{par}[x][i] = \text{par}[\text{par}[x][i-1]][i-1]$$

$$\text{par}[x][i] = \text{par}[\text{par}(x)[0]][0]$$

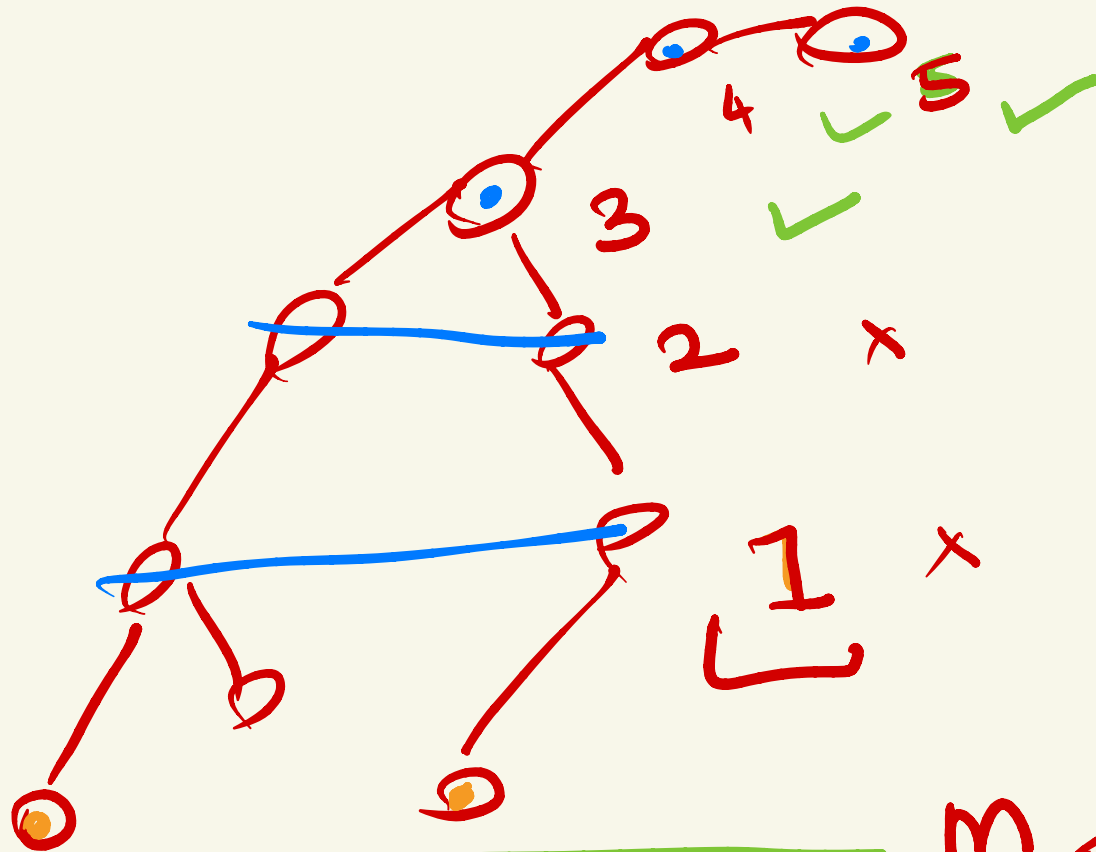
5th parent of 6



$n, q \leq 10^5$ q queries

a b
 $\hookrightarrow \text{LCA}(a, b)$

i^{th} parent of
both nodes are
equal



FFF TTT

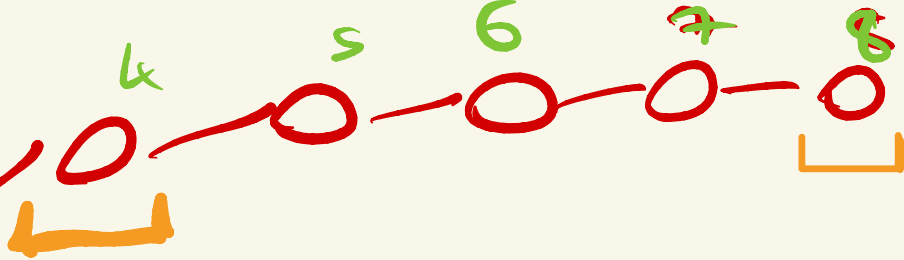
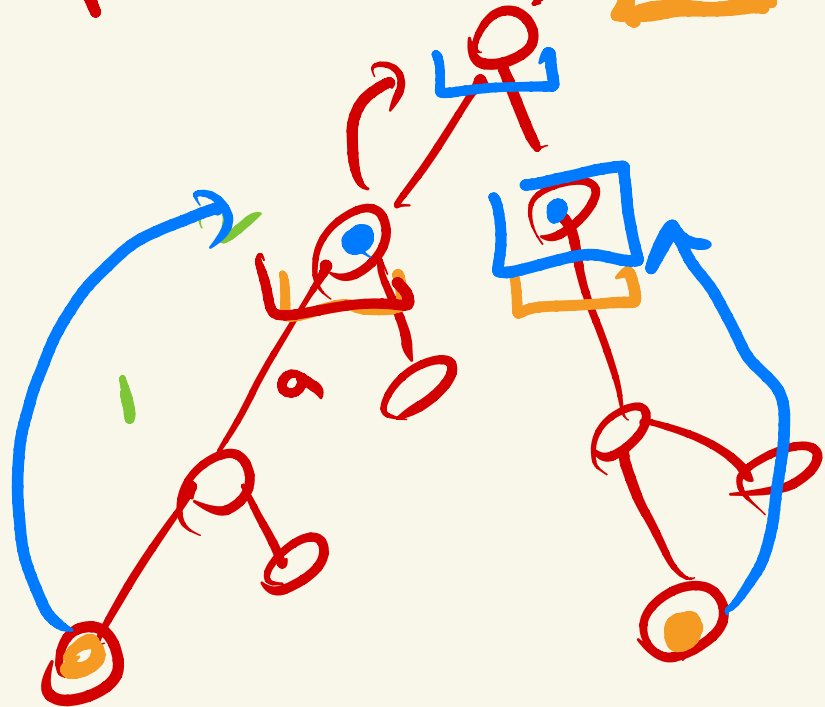
Binary
Search

$m \rightarrow$ m^{th} p of a
 $\rightarrow$ m^{th} p of b

TLE

$O(\log^2 n)$

par[a][0]



3 → 0

2^3 2^2 2^1

2^0

F F F T T T

$$2^0 + 2^1 + \dots + 2^i < 2^{i+1}$$

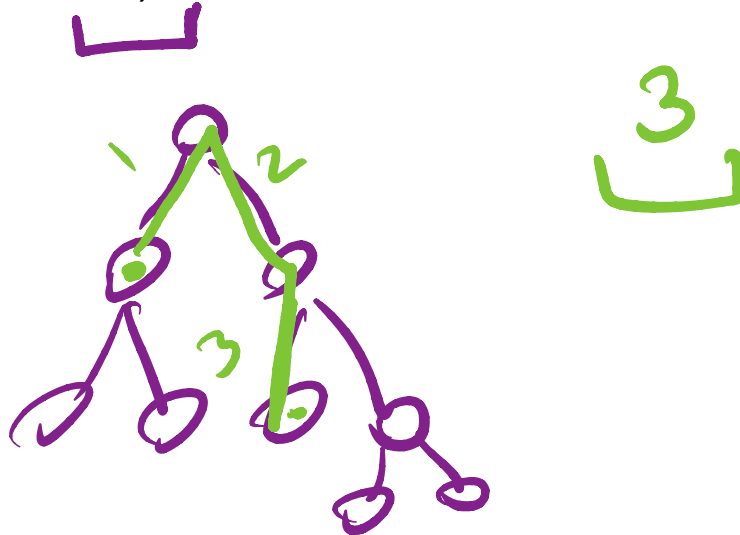
$$((i < j) \& \text{diff}) == 1$$

Distance b/w nodes Problem

Given a rooted tree with N nodes and Q queries.

For each query of the form X, Y find distance from X to Y

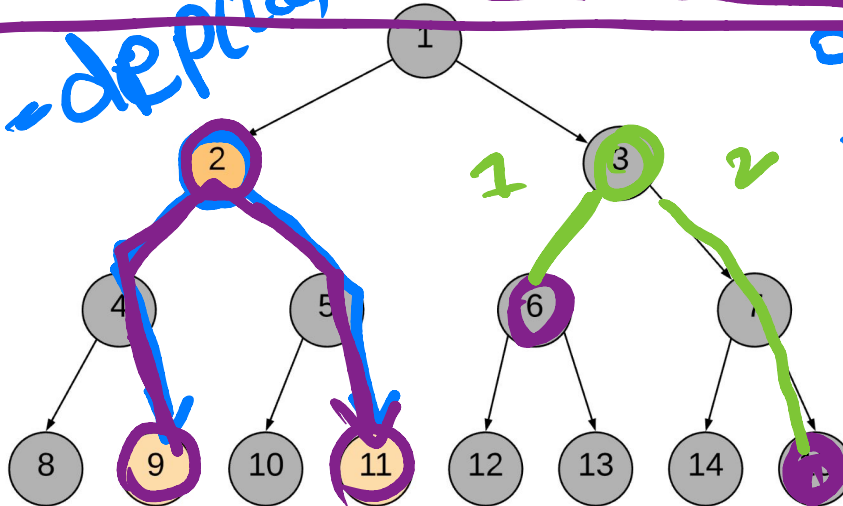
$N \leq 1e5, Q \leq 1e5$



Distance between any 2 nodes Link

$$\text{dist}(A, B) = \text{level}_A + \text{Level}_B - 2 * \text{Level}_{\text{LCA}}$$

dep(a) = depth



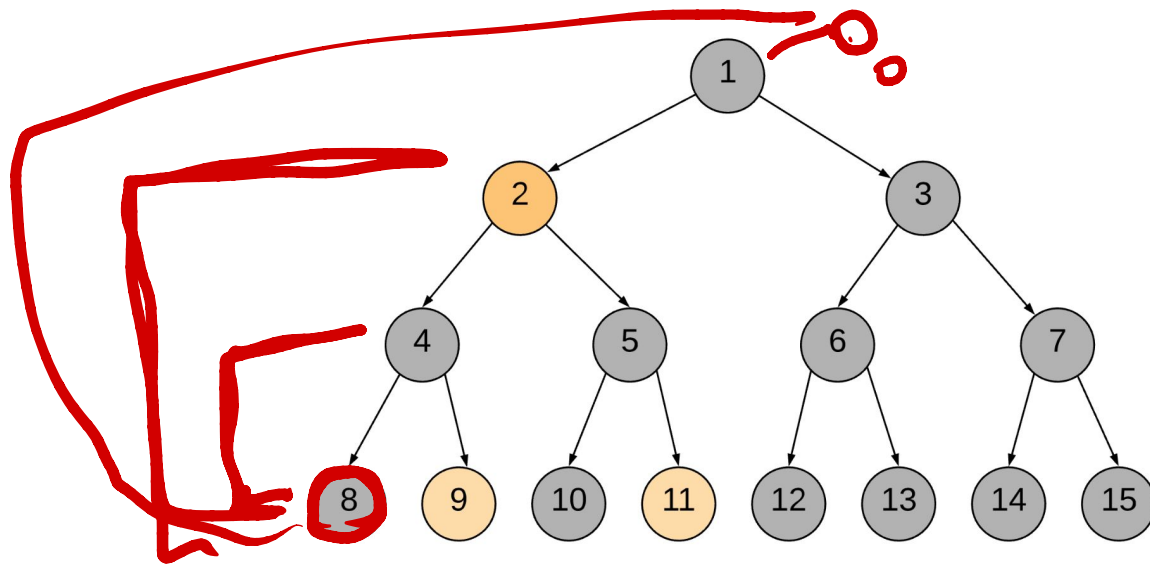
$$2 + 1 = 3$$

$$2 + 3 - 2 = 3$$

Lowest Common Ancestor for **Node 9** and **Node 11** is **Node 2**

Find minimum value on Path

$$\min(A, B) = \min(\min(A, \text{LCA}), \min(B, \text{LCA}))$$

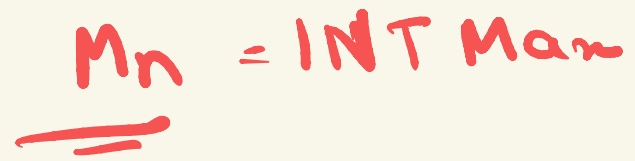


binary lifting
table

Lowest Common Ancestor for **Node 9** and **Node 11** is **Node 2**

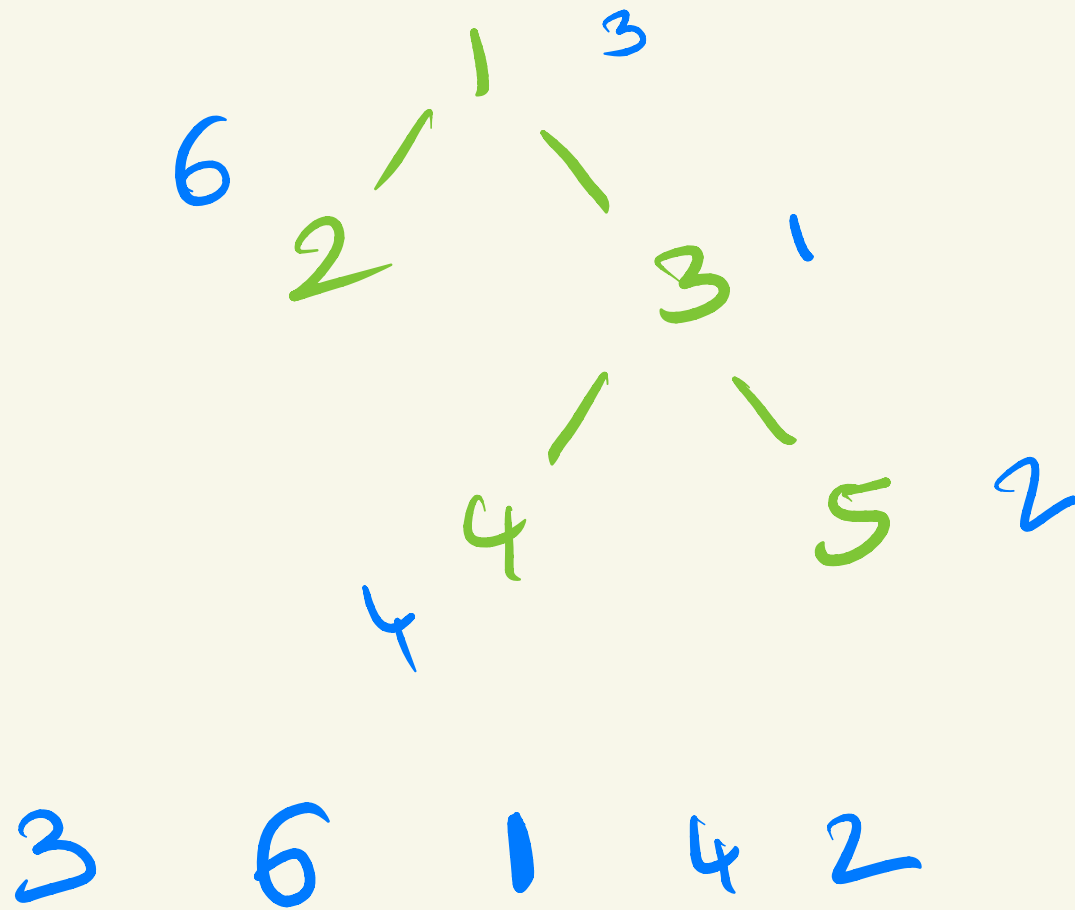
$\text{par}[\text{node}][i] =$

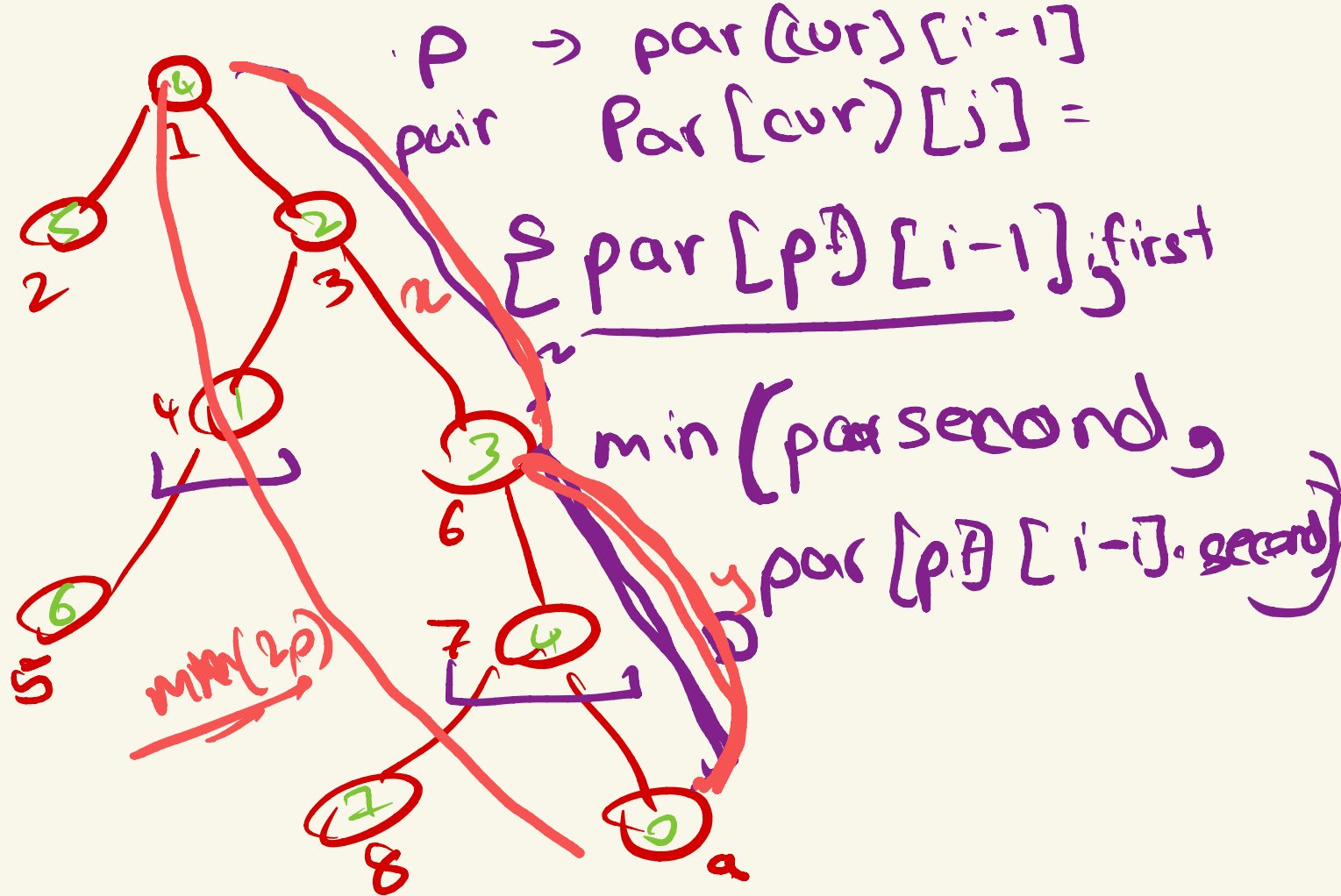
$\{ \text{2^{ith} parent, min from node to ith parent} \}$

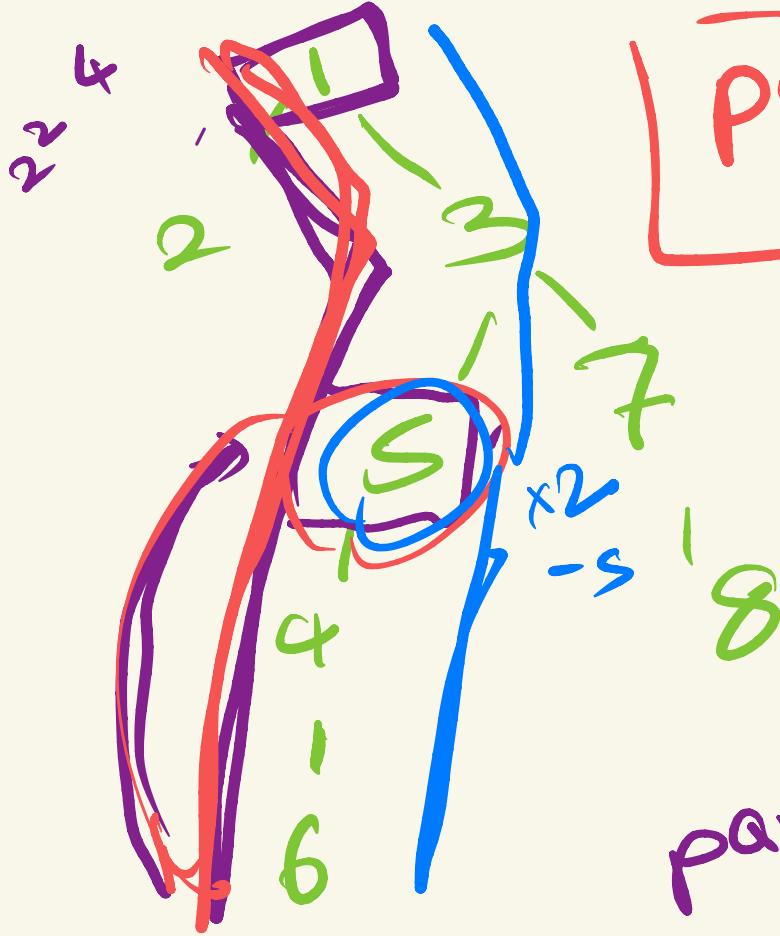


$$M_n = \min(M_n, LCA)$$

Mn -







$$\text{par}[6][2] = \{2, \min(4, 1)\}$$

$$\text{par}(5)(1) = \{1, 1\}$$

$$\text{par}(\text{node})(\text{right}) = \{5, 4\}$$